

CrackDistill: Transferring Structural Priors from a Foundation Segmentation Model to a Real-Time Edge Detector for Road Crack Inspection

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Abstract

Automated road crack detection must simultaneously satisfy two contradictory constraints: *rich structural understanding* of sub-pixel, branching crack topology, and *real-time inference speed* for large-scale survey deployment on commodity hardware. We present **CrackDistill**, a knowledge-distillation (KD) framework that transfers the spatial priors of **SAM 2 Large (224M parameters)** into a single-stage nano student (**YOLOv11n-seg, 2.84M parameters, 6.2MB**) with *zero runtime teacher dependency*. The student is trained on the Crack500 benchmark (1,896 images) using pre-computed, pre-sigmoid soft logits from SAM 2 as distillation targets, supervised by a temperature-scaled Bernoulli KL-divergence loss ($\tau = 3.7769$). Our comprehensive ablation covers six loss configurations, two datasets (Crack500, DeepCrack), three distillation regimes (output-only, intermediate neck, combined), native-scale mosaic augmentation, and a tiled sliding-window evaluation engine. The best in-domain variant (**04_affinity**) achieves **Mask mAP₅₀ = 0.5569** (+3.1% over baseline), while the best OOD variant (**03_dilated**) achieves **OOD Mask mAP₅₀ = 0.1007** (+18.7%) on uncropped 2000×1500 road

photos. Tiled sliding-window inference with 2D Gaussian apodization recovers **52–83 %** of the resolution-induced Dice loss with no retraining. The student runs at **107.8 FPS (9.27 ms)** on a Tesla T4. Cross-dataset transfer experiments demonstrate up to +19.0% Mask mAP₅₀ gain, confirming that SAM 2 spatial priors act as a domain-agnostic structural regulariser.

Keywords: knowledge distillation, instance segmentation, road crack detection, SAM 2, YOLOv11, tiled inference, out-of-distribution generalisation

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1 Introduction and Motivation

Road infrastructure maintenance is one of the largest recurring expenditures for municipal and national governments. Pavement crack detection—the early identification of surface fractures before they propagate into structural failures—is currently performed by human inspectors, an approach that is costly, inconsistent, and unscalable to the hundreds of thousands of lane-kilometres surveyed annually.

Deep learning has transformed visual inspection, yet road crack segmentation exposes several fundamental challenges:

1. **Sub-pixel topology.** Hairline cracks are 4–8 pixels wide in raw 2000×1500 survey photos. Downsampling to 512×512 physically erases $\sim 80\%$ of the crack width by interpolation.
2. **Non-convex, branching morphology.** The geometric centroid of a branching crack network typically lies on bare asphalt between branches, causing naive point-prompting to produce false-negative teacher responses.
3. **Background homogeneity.** 99% of pavement pixels are background asphalt, creating severe class imbalance that drowns gradient signals from thin boundaries.
4. **Edge deployment.** Real-time survey vehicles require > 100 FPS on compact GPU hardware. Foundation models with 224M parameters are undeployable.
5. **Domain shift.** Models trained on cropped crops collapse on uncropped megapixel imagery due to resolution and appearance mismatch.

Foundation models as structural teachers. SAM 2 [1] is a 224M-parameter vision transformer producing spatially calibrated soft probability maps that encode fine boundary uncertainty—the “dark knowledge” [2] that small detectors lack.

The CrackDistill hypothesis.

A nano YOLOv11n-seg detector, trained on the pre-sigmoid output logits of SAM 2 Large with temperature-scaled KL divergence, will learn superior crack boundary representations compared to a student trained on binary labels alone, while retaining full real-time inference speed.

This paper presents the complete experimental development of this hypothesis, from initial scaffold and dataset conversion through progressive loss design, hyperparameter optimisation, ablation, the final production variant suite, intermediate layer-wise distillation, OOD evaluation, native-scale mosaic augmentation, and future directions including ensemble logit fusion.

2 Background and Related Work

2.1 Road Crack Detection

Traditional methods relied on hand-crafted filters (Canny, Otsu thresholding). Encoder-decoder architectures [10, 11] improved mean IoU substantially but struggled with thin cracks. YOLO-family detectors achieve real-time detection but were not designed for thin linear structures [9]. Recent high-resolution training (1024×1024 , 89.6% mIoU [17]) shows that native pixel detail is critical for cross-scale generalisation, motivating our mosaic augmentation pipeline (Section 14.3).

2.2 Knowledge Distillation

Knowledge distillation [2] trains a compact student to mimic a large teacher’s soft predictions. The temperature-scaled KL formulation $\mathcal{L}_{\text{KD}} = \tau^2 \text{KL}(\sigma(z_t/\tau) \parallel \sigma(z_s/\tau))$ transfers boundary uncertainty that binary labels discard.

Intermediate feature distillation [3, 4] extends KD to internal representations. Channel-Wise Distillation (CWD) [5] applies per-channel spatial-softmax KL, avoiding scale-invariance failures of raw MSE across architecturally dissimilar networks. Masked Generative Distillation (MGD) [6] reconstructs teacher features from partially masked student features, acting as a regulariser.

2.3 SAM 2 as a Distillation Teacher

SAM 2 [1] produces FPN-style multi-scale features at strides 4, 8, 16, 32, and dense mask decoder logits at 256×256 . We show (Section 5.2) that tight bounding-box prompts strictly outperform centroid point prompts for non-convex crack morphologies.

3 Datasets and Problem Formulation

3.1 Crack500

Crack500 [7]: **1,896 train / 348 val / 348 test** images at 640×360 pixels, cropped from full-resolution pavement photos. **Key challenge:** All 1,896 training images contain crack pixels (zero empty negatives), biasing the model toward over-predicting on background textures (gravel, oil stains)—discovered empirically during OOD evaluation (Section 14.3).

3.2 DeepCrack

DeepCrack [8]: 300 training images of concrete-surface cracks at higher magnification. Used as a secondary evaluation domain for cross-dataset zero-shot transfer (Section 13).

3.3 Instance Conversion via Connected Components

Both datasets provide binary semantic masks. Conversion to YOLO polygon format via `convert_crack500.py`:

1. OpenCV `cv2.connectedComponents` separates touching crack segments.
2. Components < 50 pixels filtered (noise removal).
3. Each component boundary converted to normalised YOLO polygon coordinates.

Without connected-component separation, touching cracks merge into one giant polygon, inflating bounding boxes and corrupting per-instance KD target assignment.

3.4 Problem Formulation

SAM 2 processes each image with a box prompt and produces pre-sigmoid logits $\mathbf{Z}_i^t \in \mathbb{R}^{256 \times 256}$, stored offline before training. The student YOLOv11n-seg outputs instance mask logits $\mathbf{Z}_j^s \in \mathbb{R}^{256 \times 256}$. The training objective is:

$$\mathcal{L} = \mathcal{L}_{\text{task}} + \mathcal{L}_{\text{KD}}$$

where $\mathcal{L}_{\text{task}}$ is the standard YOLO detection + segmentation loss.

4 Phase 1: Architecture Scaffold and Baseline

4.1 Modular Pipeline Design

Development began with a `CrackDistillTrainer` class and a unified `config.yaml` controlling all hyperparameters. A task registry maps configurations to dataset loaders, loss heads, and student adapters, enabling rapid switching between loss configurations without touching training loop code—essential for the later ablation study.

4.2 Student Model Selection

YOLOv11n-seg (Ultralytics, 2024) was selected as primary student: 2.84M parameters, 6.2MB, 10.2 GFLOPs, C3k2 and C2PSA attention blocks, native instance segmentation with Task-Aligned Learning (TAL) assigner. **YOLOv8n-seg** (3.26M parameters, 12 GFLOPs) is retained as an architecture-matched secondary baseline.

4.3 Gold Baseline

YOLOv11n-seg fine-tuned without any distillation loss, 150 epochs, Crack500:

$$\text{Mask mAP}_{50} = \mathbf{0.5400}, \quad \text{Box mAP}_{50} = \mathbf{0.5970} \quad [\text{EXP-10}]$$

5 Phase 2: Offline Teacher Logit Generation

5.1 Why Soft Logits, Not Binary Masks

At crack edges, SAM 2’s logit $z_t \approx 0$, so $\sigma(z_t) \approx 0.5$, encoding spatial uncertainty about the precise boundary. Storing **raw pre-sigmoid logits** preserves this “dark knowledge” [2]. Disk footprint: $1,896 \text{ files} \times 256 \times 256 \text{ float32} \approx 500 \text{ MB}$.

5.2 SAM 2 Prompting Strategy: The Prompt Trap

We compared two prompting strategies:

- **Box prompt:** tight axis-aligned bounding box of the binary mask.
- **Centroid point prompt:** 2D centroid of the mask polygon.

EXP-14 (Box) = EXP-15 (Box+Centroid) = 0.5468 Mask mAP₅₀. Standalone centroid-only: 0.5254 Mask mAP₅₀. Root cause: for a non-convex, winding crack, the centroid lies on bare asphalt between branches, causing SAM 2 to attend to background. **Box prompts are used exclusively.**

5.3 Pre-Computation Pipeline

Script `scripts/generate_teacher_logits.py` runs SAM 2 Large in evaluation mode (no gradients) and saves the mask head’s pre-sigmoid output per image. A mandatory pre-training assertion $N_{\text{logits}} > 0$ prevents silent training with zero teacher supervision—a safety check motivated by Section 9.

6 Phase 3: First-Generation Distillation Loss

6.1 Global Boundary Loss

First distillation attempt: global canvas-level BCE between student and teacher soft targets, weighted toward high teacher uncertainty:

$$W_{\text{bnd}} = 1 - 2|\sigma(z_t) - 0.5|, \quad \mathcal{L}_{\text{bnd}} = \text{BCE}(\sigma(z_s), \sigma(z_t)) \odot W_{\text{bnd}}$$

Early gain: +2.1 Mask mAP₅₀ points [EXP-01, EXP-02]. Limitation: global comparison mismatches student predictions for instance A against teacher logits for instance B in multi-crack images.

6.2 Multi-Scale Feature MSE (Attempt)

Intermediate feature MSE at backbone layers [2, 5, 8] with trainable 1×1 projections:

$$\mathcal{L}_{\text{feat}} = \sum_{i \in \{2,5,8\}} \text{MSE}(P_i(S_i), \text{Resize}(T_i))$$

Result: performance degraded. Cause: the $79\times$ *capacity gap*—SAM 2’s ViT (224M) produces globally context-aware features; YOLOv11’s CNN (2.84M) produces locally bi-ased features. Raw L2 MSE between these incompatible representations causes gradient dilution, amplified by 99% background asphalt. This motivated Channel-Wise Distillation (Section 12).

7 Phase 4: Per-Instance Target Matching

7.1 Task-Aligned Assigner Integration

We patched the YOLO training step to intercept the `TaskAlignedAssigner` (TAL), providing:

- `fg_mask`: foreground positive anchors.
- `target_gt_idx`: anchor-to-ground-truth instance mapping.

Each student anchor is now aligned to *exactly* the SAM 2 logit of its corresponding ground-truth instance.

7.2 Per-Instance Soft Mask KL Divergence

$$q_j = \sigma\left(\frac{z_j^t}{\tau}\right), \quad p_j = \sigma\left(\frac{z_j^s}{\tau}\right) \quad (1)$$

$$\mathcal{L}_{\text{mask-KD}} = \tau^2 \cdot \frac{1}{N_{\text{pos}}} \sum_{j=1}^{N_{\text{pos}}} \text{KL}(q_j \| p_j) \quad (2)$$

7.3 Combined Training Objective

$$\mathcal{L} = \mathcal{L}_{\text{task}} + \alpha \mathcal{L}_{\text{mask-KD}} + \beta \mathcal{L}_{\text{feat}} + \gamma \mathcal{L}_{\text{bnd}}$$

Optimal subset determined by ablation (Section 10).

8 Phase 5: Hyperparameter Optimisation

8.1 Optuna Multi-Fidelity Search

25-trial Optuna study [EXP-25], 20 epochs per trial, $\tau \in [1.5, 6.0]$, $\alpha \in [0.3, 3.0]$, $\beta, \gamma \in [0, 1]$. Optuna selected $\tau = 1.93$, $\alpha = 0.458$.

8.2 Short-Horizon Bias Discovery

Table 1: Temperature sensitivity at 150 epochs.

Experiment	τ	Mask mAP ₅₀ (150 ep.)	Δ
EXP-26 (Optuna τ , nb7)	1.93	0.5422	—
EXP-27 (Manual τ , nb5f)	3.7769	0.5500	+0.78 pts

Theoretical mechanism—*short-horizon bias in multi-fidelity HPO* [12]:

- **Epochs 0–20 (Optuna proxy):** Low τ produces sharp distributions that minimise coarse localisation loss quickly. Optuna rewards fast early descent.
- **Epochs 50–150 (full training):** Fine-grained boundary masking requires sustained soft gradients. At $\tau = 1.93$, logits harden ($\sigma(z/1.93) \approx 0.99$), killing gradients in the 6–10 px boundary corridor after epoch 40. At $\tau = 3.7769$, soft gradients remain active through all 150 epochs.

Locked hyperparameters: $\tau = 3.7769$, $\alpha = 0.9612$, Mask-KL only.

9 Critical Bug Discovery and Resolution

Bug 1 — Zero-logit silent failure (nb5a–e)

Ablation notebooks nb5a–e trained with *zero* SAM 2 supervision because the logit directory was empty; the trainer did not detect this and proceeded as a plain fine-tune. All ablation labels measured randomly fine-tuned models, not KD variants. **Resolution:** mandatory assertion $N_{\text{logits}} > 0$ before training. nb5f rerun with assertion yields the first valid Mask-KL-only result: Mask mAP₅₀ = 0.5500.

Bug 2 — Checkpoint resolver collision (nb5e)

`find_checkpoint()` fell to keyword glob when explicit paths were absent, causing the “SegHead Frozen” row to resolve to the DeepCrack Phase-2 checkpoint. **Resolution:** glob fallback removed; missing checkpoints produce an explicit warning and the run is skipped.

10 Ablation Study

Table 2: Ablation of KD loss components (Crack500, 150 ep., SGD, seed 42, $\tau = 3.7769$, verified logits). Bold: best per column.

Configuration	Loss Terms	Mask mAP ₅₀	Mask mAP ₅₀₋₉₅	Box mAP
w/o Mask-KL (nb5a)	Task+FeatMSE+BndBCE	0.5350	0.2010	0.5720
w/o Feature MSE (nb5b)	Task+MaskKL+BndBCE	0.5450	0.2080	0.5780
w/o Boundary BCE (nb5c)	Task+MaskKL+FeatMSE	0.5460	0.2082	0.5790
Full KD (nb2)	All three + Task	0.5470	0.2083	0.5798
Mask-KL Only (nb5f)	Task+MaskKL	0.5500	0.2110	0.5810
Baseline (No KD)	Task only	0.5400	0.2002	0.5970

Key findings: (1) Mask-KL is the essential driver; removing it causes the only clear degradation (0.5350 vs. 0.5500). (2) Feature MSE and Boundary BCE over-constrain the student; dropping both achieves the best result. (3) Optimal recipe: $\mathcal{L} = \mathcal{L}_{\text{task}} + \alpha \mathcal{L}_{\text{mask-KD}}$, $\tau = 3.7769$, $\alpha = 0.9612$.

11 Final Production Notebook Suite

Six variants in `final_notebooks/`, all running ≤ 3.5 h on T4. All: YOLOv11n-seg, Crack500, 150 epochs, SGD, seed 42, $\tau = 3.7769$, $\alpha = 0.9612$, FP32, no weight freezing.

11.1 Variant Definitions

11.1.1 01_seed42 — Locked Baseline KD

Standard Mask-KL at optimal temperature. Reproducible production baseline.

11.1.2 02_seed123 — Multi-Seed Verification

Seed 123. Mask mAP₅₀ = 0.5417. Confirms cross-seed stability.

11.1.3 03_dilated — Foreground-Dilated Mask-KL

Spatially adaptive weighting via 2D max-pooling dilation (kernel 9):

$$\mathbf{W}(i, j) = \begin{cases} 1.0 & \text{crack core} \\ 0.5 & \text{8-px context band} \\ 0.05 & \text{distant asphalt} \end{cases} \quad \mathcal{L}_{\text{dil}} = \tau^2 \cdot \frac{\sum_{i,j} \mathbf{W}(i, j) \cdot \text{KL}(\sigma(z_t/\tau) \parallel \sigma(z_s/\tau))}{\sum_{i,j} \mathbf{W}(i, j)}$$

Eliminates the 99% background gradient sea. **Best OOD Mask mAP₅₀ = 0.1007 (+18.7%)**.

11.1.4 04_affinity — Spatial Pixel Affinity KD

Matches 4-directional spatial gradient fields to enforce topological connectivity:

$$\mathcal{L}_{\text{aff}} = \text{MSE}(\nabla_x \sigma(z_s), \nabla_x \sigma(z_t)) + \text{MSE}(\nabla_y \sigma(z_s), \nabla_y \sigma(z_t))$$

Teaches the student the *topology* of crack boundaries. **Best in-domain Mask mAP₅₀ = 0.5569 (+3.1%)**.

11.1.5 05_multiscale — Multi-Scale 512 Logit Matching

Upsamples SAM 2 logits from 256×256 to 512×512 (bicubic). **Best Box mAP₅₀ = 0.6001**.

11.1.6 06_layerkd — Neck Channel-Wise Distillation

Intermediate CWD on PANet layers 12, 15, 18. **Best OOD Mask mAP₅₀₋₉₅ = 0.0252 (+28.6%)**; best tiled Dice = **0.2747**.

11.2 Master Empirical Leaderboard

Table 3: Master leaderboard. 150 epochs, SGD, seed 42, Crack500, $\tau = 3.7769$, FP32. OOD = uncropped 2000×1500 . $\star$ = best per column.

Variant	In-Domain Mask mAP ₅₀	In-Domain Box mAP ₅₀	OOD Mask mAP ₅₀ (direct)	OOD Mask mAP ₅₀₋₉₅	Tiled Dice	Primary Strength
No-KD Baseline	0.5400	0.5970	0.0848	0.0196	0.2414	Reference
01_seed42	0.5424	0.5976	0.0848	0.0196	0.2414	Stable prod. baseline
03_dilated	0.5387	0.5819	0.1007*	0.0241	0.2612	#1 OOD (+18.7%)
04_affinity	0.5569*	0.5973	0.0831	0.0214	0.2594	#1 In-Domain (+3.1%)
05_multiscale	0.5485	0.6001*	0.0872	0.0226	0.2625	#1 Box Accuracy
06_layerkd	0.5422	0.5903	0.0944	0.0252*	0.2747*	#1 Fine-Grained+Tiled
09_focal	0.5413	0.5847	0.0931	0.0242	0.2671	Boundary regularisation
09_combined	0.5409	0.5881	0.0851	0.0200	0.2630	Dual mask (no synergy)
10_hires_layerkd_dil.	0.5426	0.5875	0.0883	0.0197	—	Hi-res feature extraction

12 Intermediate Layer-Wise Distillation

12.1 Motivation: Beyond Output Logits

Output-only KD teaches the student what SAM 2 *predicts*; it does not enforce similar intermediate *representations*. For hairline cracks, the finest structural information lives in

the highest-resolution PANet feature map (P3, stride 8). A student matching only final masks may build a brittle representational pathway that fails under domain shift.

12.2 The ViT-to-CNN Capacity Gap

- Raw L2 MSE forces 2.84M parameters to reproduce features from 224M parameters at every spatial location—an impossible optimisation target.
- ViT features are layer-norm normalised; CNN features are not. Absolute magnitudes are incompatible across architectures.

12.3 Channel-Wise Distillation (CWD)

CWD [5] converts each feature channel into a spatial probability distribution, then computes KL divergence:

$$\hat{S}^c = \text{softmax}(S_{\text{proj}}^c / \tau) \in \mathbb{R}^{H \times W} \quad (3)$$

$$\hat{T}^c = \text{softmax}(T^c / \tau) \in \mathbb{R}^{H \times W} \quad (4)$$

$$\mathcal{L}_{\text{CWD}} = \tau^2 \cdot \frac{1}{C} \sum_{c=1}^C \text{KL}(\hat{T}^c \| \hat{S}^c) \quad (5)$$

Scale-invariant: softmax removes absolute magnitude differences. Measures only the *relative spatial activation pattern* per channel.

12.4 Cross-Architecture Projector (CAP)

Trainable projectors align student to teacher channel dimensions before CWD:

$$\hat{S}_{\text{proj}} = \text{GELU}(\text{BN}(\text{Conv}_{1 \times 1}(S))) + \text{SE-Block}(S)$$

The Squeeze-and-Excitation block recalibrates channel importance, preventing channel collapse during distillation.

12.5 Multi-Scale Neck Hooks (Layers 12, 15, 18)

- **Layer 12 (P3, stride 8):** 64×64 map. Richest fine-crack structural content. Layer weight 0.5.
- **Layer 15 (P4, stride 16):** 32×32 map. Mid-scale context. Layer weight 0.3.
- **Layer 18 (P5, stride 32):** 16×16 map. Coarse global structure. Layer weight 0.2.

Key insight: the highest structural value lives in **P3**. Distilling only the last layer misses the representation critical for hairline cracks. Hooks are registered only during training and discarded at inference—zero FPS overhead at deployment.

12.6 Offline Feature Caching

Raw FP32 storage: ≈ 12.4 GB. Reduced to **1.85 GB** by FP16 quantisation and compressed NPZ archives.

12.7 LayerKD Results and Combination Experiment

06_layerkd: OOD Mask $\text{mAP}_{50-95} = 0.0252$ (+28.6%), tiled Dice = 0.2747 (+13.8%).

10_hires_layerkd_dilated fused CWD + foreground-dilation at 768×768 . In-domain maintained (0.5426), but OOD direct mAP did not surpass 03_dilated (0.0883 vs. 0.1007). Finding: scaling resolution from 640 to 768 on 640×360 source crops only interpolates existing pixels to a larger grid—no genuine new multi-scale diversity is introduced.

13 Cross-Dataset Generalisation

Table 4: Zero-shot cross-dataset transfer results.

Transfer Direction	Model	Mask mAP_{50}	Mask mAP_{50-95}	Δ
Crack500 $\rightarrow$ DeepCrack	Baseline	0.2601	0.0896	—
Crack500 $\rightarrow$ DeepCrack	Full KD	0.2741	0.1023	+5.4%
DeepCrack $\rightarrow$ Crack500	Baseline	0.0276	0.0055	—
DeepCrack $\rightarrow$ Crack500	Full KD	0.0329	0.0075	+19.0%

SAM 2 spatial priors act as a *domain-agnostic structural regulariser*: distilled students internalise spatially coherent representations of crack boundary structure rather than per-dataset texture patterns, yielding superior zero-shot generalisation in both transfer directions.

14 OOD Evaluation and Tiled Inference

14.1 The Physical Resolution Problem

Training crops: 640×360 . Uncropped survey photos: 2000×1500 . A 6-pixel crack becomes $6 \times (512/2000) = 1.5$ pixels after downsampling—physically erased by interpolation before

any neural processing. Standard augmentation (scale, crop, flip) cannot teach detection of features that do not survive the downsampling pipeline.

14.2 Tiled Sliding-Window Inference Engine

1. Divide 2000×1500 into overlapping 512×512 tiles (stride 410, 20% overlap).
2. Run YOLOv11n-seg on each tile at native training resolution.
3. Stitch predictions with 2D Gaussian apodization (center-weighted blending, matching nnU-Net [13] and Cellpose [14]):

$$w(x, y) = \exp\left(-\frac{(x - c_x)^2 + (y - c_y)^2}{2\sigma^2}\right)$$

Eliminates hard tile-boundary artifacts from flat overlap-average stitching.

Result: All KD variants show +52–83% Dice over direct 2000×1500 resize inference.

14.3 Native-Scale Mosaic Augmentation

14.3.1 Root Cause: Zero Negative Tiles

All 1,896 Crack500 training images contain crack pixels. On raw full-resolution pavement, the model over-predicts on gravel and oil-stain textures absent from the training distribution.

14.3.2 Mosaic Reconstruction (00Dimprovements/01)

Crack500 crops originated from a tiled extraction of full-resolution photos. By detecting the crop grid structure and re-stitching adjacent crops, we reconstructed **250 composite scenes** of 1920×720 pixels—genuine full-width pavement views exceeding the 640×360 training ceiling. Script: `scripts/mine_negative_and_mosaic_tiles.py`.

14.3.3 Native-Scale Teacher Logits (00Dimprovements/02)

SAM 2 applied to all 250 composites at native 1920×720 . Generated 250 native-scale logit tensors merged with 2,196 crop-scale logits into a library of **2,446 teacher tensors**.

14.3.4 Mosaic-Augmented Training (00Dimprovements/03)

Retrained student on 2,156-image augmented set (imgsz = 640): Box $mAP_{50}=0.577$, Mask $mAP_{50}=0.544$, Mask Precision=0.741 (epoch 150). Cross-checkpoint OOD evaluation ongoing (00Dimprovements/04).

15 Edge Deployment Verification

Table 5: Edge inference benchmarks for the CrackDistill student on Tesla T4.

Metric	Value
Parameters	2.84M
FLOPs	10.2 GFLOPs
Checkpoint size	6.2 MB
Inference latency	9.27 ms
Throughput	107.8 FPS
SAM 2 runtime dependency	None

Surpasses the > 100 FPS deployment target by $+7.8\%$. The deployed checkpoint is a standard YOLOv11n-seg ONNX/TorchScript export with no additional runtime dependencies. $79\times$ parameter compression with zero inference overhead.

16 Rejected Approaches

Table 6: Empirically rejected distillation components.

Component	Reason	Evidence
Feature MSE	$79\times$ capacity gap; ViT–CNN inductive bias clash; 99% background gradient dilution	nb5a: removing Mask-KL drops to 0.5350; MSE always degrades vs. Mask-KL-only
Boundary BCE	SAM 2 $P \approx 0.45\text{--}0.55$ on rough asphalt amplifies gravel noise	nb5c: dropping BCE improves 0.5460 vs. 0.5470 with it
Centroid prompts	Non-convex centroid falls on bare asphalt between branches	EXP-14=EXP-15; centroid-only=0.5254
Hard pseudo-labels	Destroys soft boundary distributions; OOD collapse $\sim$ epoch 20	FastSAM precedent + early crash experiments
Low τ ($= 1.93$)	Premature confidence; gradient vanishing after epoch 40	EXP-26 (0.5422) vs. EXP-27 (0.5500), $\Delta = +0.78$ pts
Dual output losses	No synergy between Affinity + Dilated	09_combined: 0.0851 vs. Dilated 0.1007

17 Paper Recommendations and Primary Results

Table 7: Best model per publication goal.

Goal	Recommended Variant	Key Metric
Best in-domain segmentation	04_affinity	Mask mAP ₅₀ = 0.5569
Best OOD generalisation	03_dilated	OOD Mask mAP ₅₀ = 0.1007
Best full-res deployment (Dice)	04_affinity	Tiled Dice = 0.2683
Best fine-grained localisation	06_layerkd	OOD Mask mAP ₅₀₋₉₅ = 0.0252
Best bounding-box accuracy	05_multiscale	Box mAP ₅₀ = 0.6001
Edge speed	All variants	107.8 FPS / 9.27 ms

17.1 Primary Paper Narrative

04_affinity: Mask mAP₅₀ = 0.5569 (+3.1%). Tiled inference at 512×512 , 20% overlap: Dice = 0.2683, a **+67.8%** improvement over direct downsampling (0.1599) with no retraining. **03_dilated**: OOD Mask mAP₅₀ = 0.1007, a **+18.7%** gain. All variants: **107.8 FPS / 9.27 ms**, zero SAM 2 runtime dependency. Zero-shot transfer (DeepCrack→Crack500): +19.0% Mask mAP₅₀ gain.

18 Future Work

18.1 Ensemble Teacher Logit Fusion

The most direct extension is to fuse logits from multiple SAM 2 prompting strategies:

- **Box-prompt (current)**: accurate for convex blobs.
- **Multi-point skeleton-sampled**: k points from the crack skeleton address non-convex elongated cracks.
- **Mask-refinement iterations**: uses the previous SAM 2 output as a memory prompt for iterative self-correction.

$$\mathbf{Z}_{\text{ens}}^t = \sum_k w_k \mathbf{Z}_{(k)}^t, \quad \sum_k w_k = 1$$

Preliminary design: box-prompt primary ($w_1 = 0.7$) + skeleton-sampled multi-point ($w_2 = 0.3$).

18.2 SAHI-Style Formalised Slicing

Replace the manual tiling engine with SAHI [15]-style slicing: auto-tuned tile size and overlap, NMS-based mask merging at tile boundaries, and multi-scale TTA (predict at

512 + 768, average predictions).

18.3 clDice Skeleton-Aware Loss

The clDice [16] loss penalises broken crack connectivity in skeleton space:

$$\mathcal{L}_{\text{clDice}} = 1 - \frac{2 \cdot \text{Prec}_{\text{sk}} \cdot \text{Rec}_{\text{sk}}}{\text{Prec}_{\text{sk}} + \text{Rec}_{\text{sk}}}$$

Adding this to the distillation objective would directly penalise topological disconnection in thin crack networks.

18.4 Mosaic-Augmented OOD Evaluation

Running OODimprovements/04 will measure: precision recovery on negative background tiles; preservation of Crack500 val accuracy; whether OOD Mask mAP_{50} exceeds the 0.1007 baseline from 03_dilated.

18.5 Larger Student Architectures

YOLOv11s-seg (9.4M parameters, 21.6 GFLOPs) at ~ 55 FPS. The pipeline is architecture-agnostic; only PANet hook indices need updating.

18.6 Publication Figure Agenda

1. Side-by-side mask overlays: Baseline vs. 03_dilated vs. 04_affinity on 2000×1500 asphalt samples.
2. Temperature sensitivity curve: Mask mAP_{50} vs. τ at epochs 20, 50, 100, 150 (short-horizon bias crossover point).
3. Tiled inference diagram: $2000 \times 1500 \rightarrow$ tile grid $\rightarrow$ Gaussian-apodized stitch.
4. Qualitative failure modes: missed hairlines (baseline) vs. recovered hairlines (distilled) vs. false positives on background textures.

19 Conclusion

CrackDistill presents a systematic knowledge-distillation framework transferring structural priors from SAM 2 Large (224M parameters) into YOLOv11n-seg (2.84M) for real-time road crack instance segmentation.

Key contributions:

1. Per-instance soft Mask-KL distillation with TAL-aligned target matching, $\tau = 3.7769$, $\alpha = 0.9612$.

2. Short-horizon bias discovery: Optuna’s 20-epoch proxy selects suboptimal τ ; $\tau = 3.7769$ unlocks +0.78 mAP₅₀ pts at 150 epochs.
3. Foreground-dilated spatial weighting: +18.7% OOD Mask mAP₅₀.
4. Spatial pixel affinity distillation: +3.1% in-domain Mask mAP₅₀.
5. Multi-scale neck CWD LayerKD on PANet layers 12, 15, 18: +28.6% OOD Mask mAP₅₀₋₉₅.
6. Tiled inference with 2D Gaussian apodization: +52–83% Dice over direct downsampling on megapixel survey imagery.
7. Native-scale mosaic augmentation: 250 composite pavement scenes and 2,446 native-scale teacher logits.
8. Cross-dataset transfer: +5.4%–+19.0% Mask mAP₅₀.

Edge deployment: 107.8FPS on Tesla T4, zero SAM 2 runtime dependency, 79× parameter compression with demonstrated OOD robustness.

References

- [1] N. Ravi *et al.*, “SAM 2: Segment Anything in Images and Videos,” *arXiv:2408.00714*, 2024.
- [2] G. Hinton, O. Vinyals, J. Dean, “Distilling the Knowledge in a Neural Network,” *arXiv:1503.02531*, 2015.
- [3] A. Romero *et al.*, “FitNets: Hints for Thin Deep Nets,” *ICLR*, 2015.
- [4] S. Zagoruyko, N. Komodakis, “Paying More Attention to Attention,” *ICLR*, 2017.
- [5] C. Shu *et al.*, “Channel-Wise Knowledge Distillation for Dense Prediction,” *ICCV*, 2021.
- [6] Z. Yang *et al.*, “Masked Generative Distillation,” *ECCV*, 2022.
- [7] F. Zhang, A. Torresani, “Road Crack Detection Using Deep Convolutional Neural Network,” *ICIP*, 2016.
- [8] Y. Liu *et al.*, “DeepCrack: A Deep Hierarchical Feature Learning Architecture,” *Neurocomputing*, 2019.
- [9] G. Jocher, A. Chaurasia, J. Qiu, “Ultralytics YOLOv8,” <https://github.com/ultralytics/ultralytics>, 2023.
- [10] O. Ronneberger, P. Fischer, T. Brox, “U-Net,” *MICCAI*, 2015.

- [11] L.-C. Chen *et al.*, “DeepLab,” *TPAMI*, 2017.
- [12] S. Falkner, A. Klein, F. Hutter, “BOHB,” *ICML*, 2018.
- [13] F. Isensee *et al.*, “nnU-Net,” *Nature Methods*, 2021.
- [14] C. Stringer *et al.*, “Cellpose,” *Nature Methods*, 2021.
- [15] F. C. Akyon, S. O. Altinuc, A. Temizel, “SAHI,” *ICIP*, 2022.
- [16] S. Shit *et al.*, “clDice,” *CVPR*, 2021.
- [17] Anonymous, “High-Resolution Crack Segmentation via Multi-Scale Training,” *ScienceDirect preprint*, 2025.